

Principles of Finance

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MSc Climate Change, Management and Finance

Week 3



Module Overview

Week 1

Introduction, Institutions, Financial Decision making, Time value of money, Net Present Value (NPV) Rule, Cost-Benefit Analysis (**BD 1-3**)*

Week 2

Time-Value of Money (Perpetuities/Annuities), Interest Rates, Valuing Bonds (**BD 4-6**)*

Week 3

Investment Decision Rules, Fundamentals of Capital Budgeting, Valuing Stocks (**BD 7.1, 7.2 (first part), 8, 9**)* , **Saito Solar case**

Deadline for submission of group assignment 1 is 9am, Thursday, 13th November

Week 4

Capital Markets and Pricing of Risk (**BD 10**), Optimal Portfolio Choice and the CAPM (**BD 11 excl. Appendix**), Estimating the Cost of Capital (**BD 12.1-13.3, excl. Appendix**)

Week 5

Capital Structure, Debt and Taxes (**BD 12.4, 12.5, 12.6, 12.7, BD 14.1, 14.2, 14.3**), Investor Behaviour and Capital Market Efficiency (**BD 13.1, 13.2, 13.6-13.8 excl. Appendix**), Introduction to the balance sheet and asset-liability-management (ALM) of a bank, insurance company and pension fund, **M&M Pizza Case**

*(See **reading list** for detailed chapter references and what is excluded as well as additional optional readings)

Deadline for submission of group assignment 2 is 9am, Tuesday, 2nd December

Imperial means Intelligent Business

This week develops
cash-flow-based
valuation skills for
investment decisions

Week 3 Lectures Learning objectives

BD Chapter 7

- Internal Rate of Return rule for simple cases (negative cash flows first, same risk,...) *

BD Chapter 8

- Given a set of facts, identify relevant cash flows for a capital budgeting problem.
- Explain why opportunity costs must be included in cash flows, while sunk costs and interest expense must not.
- Calculate free cash flows for a given project.
- Illustrate the impact of depreciation expense on cash flows.

BD Chapter 9

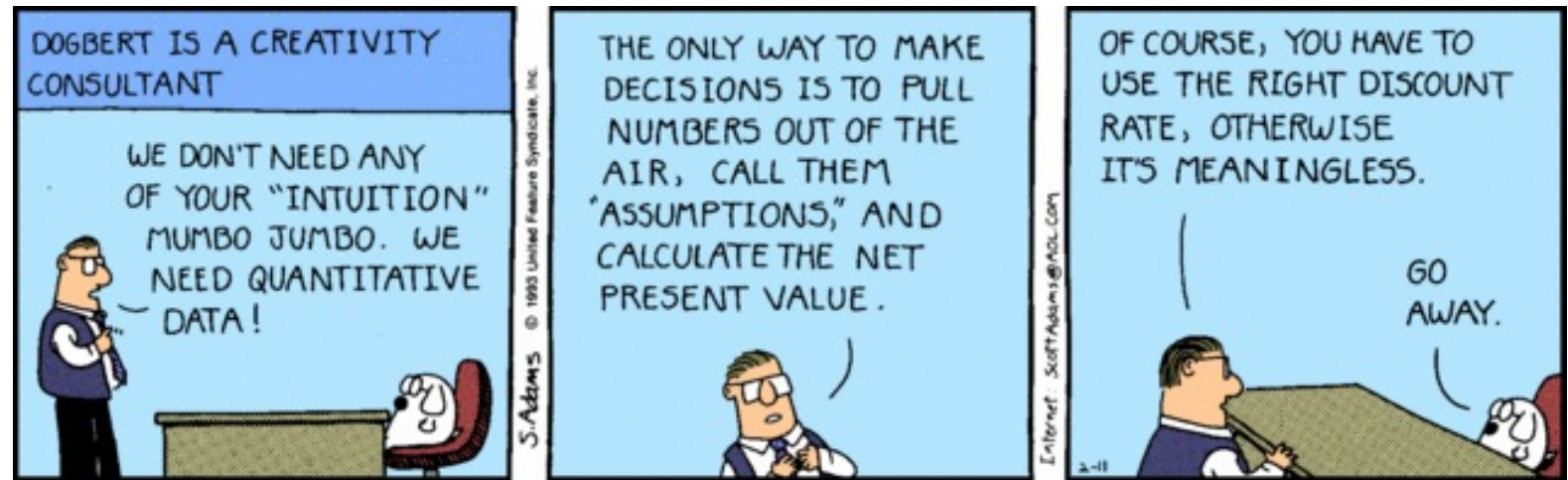
- Use the dividend-discount model to compute the value of a dividend-paying company's stock, whether the dividends grow at a constant rate starting now or at some time in the future.
- Use the discounted free cash flow model to calculate the value of stock in a company with leverage.
- Use comparable firm multiples to estimate stock value.
- Explain why several valuation models are required to value a stock.

* (**Please note** that Payback Rule and complex IRR cases (that is BD Chapters 7.3-7.5) will not be covered in the interest of time)

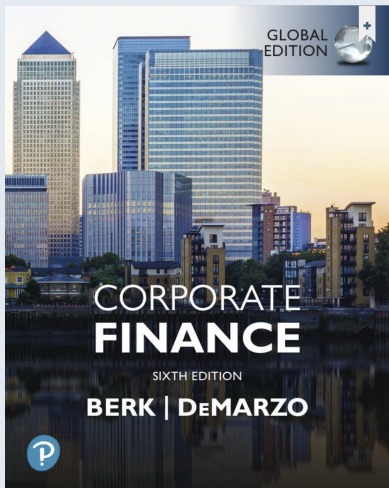
Motivating examples



On a lighter note



Chapter 7 (BD)



Chapter Outline

7.1 NPV and Stand-Alone Projects

7.2 The Internal Rate of Return Rule

NPV captures value creation by discounting all future cash flows

7.1 NPV and Stand-Alone Projects

- Consider a take-it-or-leave-it investment decision involving a single, stand-alone project for Fredrick's Feed and Farm (FFF).
 - The project costs \$250 million and is expected to generate cash flows of \$35 million per year, starting at the end of the first year and lasting forever.
- The NPV of the project is calculated as:

$$NPV = -250 + \frac{35}{r}$$

- The NPV is dependent on the discount rate.

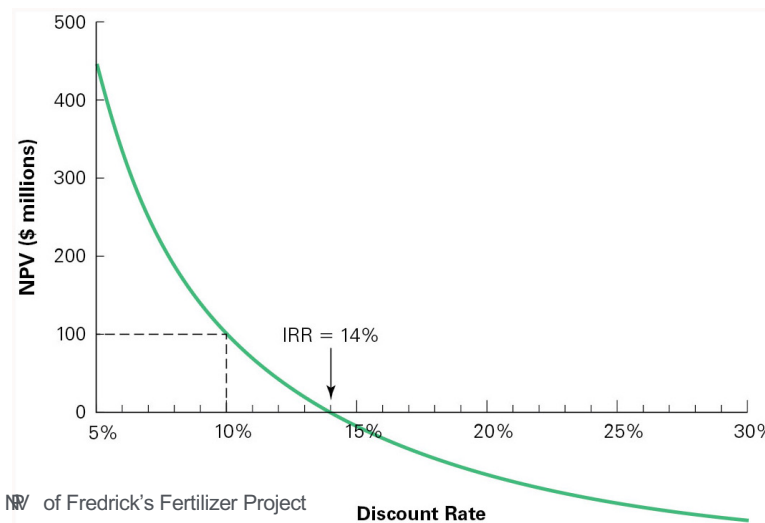


Figure 7.1 NPV of Fredrick's Fertilizer Project

If FFF's cost of capital is 10%, the NPV is \$100 million, and they should undertake the investment

When in doubt,
follow NPV — it
always maximizes
value

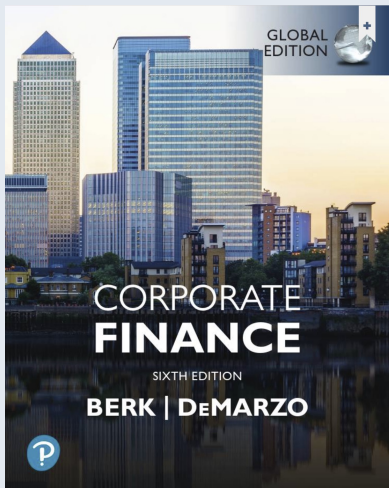
7.2 The Internal Rate of Return Rule

Simple case

Complex cases and
disagreement

- Sometimes alternative investment rules may give the same answer as the NPV rule, but at other times they may disagree.
 - When the rules conflict, the NPV decision rule should be followed.
- **Internal Rate of Return (IRR) Investment Rule**
 - Take any investment where the IRR exceeds the cost of capital
 - Turn down any investment whose IRR is less than the cost of capital
- In general, the IRR rule works for a stand-alone project if all of the project's negative cash flows precede its positive cash flows.
 - In Figure 7.1, whenever the cost of capital is below the IRR of 14%, the project has a positive NPV, and you should undertake the investment.
- In other cases, the IRR rule may disagree with the NPV rule and thus be incorrect.
 - Delayed Investments
 - Nonexistent IRR
 - Multiple IRRs

Chapter 8 (BD)



Chapter Outline

- 8.1** Forecasting Earnings
- 8.2** Determining Free Cash Flow and NPV
- 8.3** Choosing Among Alternatives
- 8.4** Further Adjustments to Free Cash Flow
- 8.5** Analyzing the Project

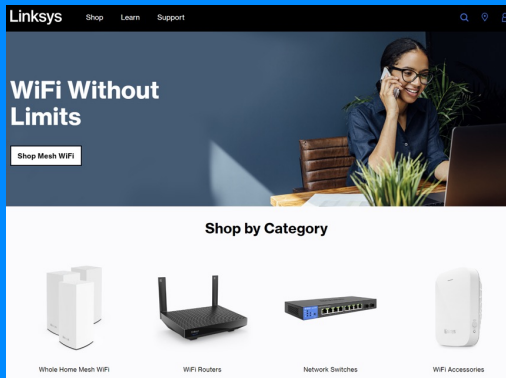
Valuation begins
with realistic
forecasts of
incremental
earnings

8.1 Forecasting Earnings

- Capital Budget
 - Lists the investments that a company plans to undertake
- Capital Budgeting
 - Process used to analyze alternate investments and decide which ones to accept
- Incremental Earnings
 - The amount by which the firm's earnings are expected to change as a result of the investment decision

Example: Cisco and Homenet

Revenue and Cost Estimates



- Example
 - Cisco has completed a \$300,000 feasibility study to assess the attractiveness of a new product, HomeNet.
 - The project has an estimated life of **4 years**.
 - **Revenue Estimates**
 - Sales = 100,000 units/year
 - Per Unit Price = \$260so \$26,000,000 per year
 - **Cost Estimates**
 - Up-Front R & D = \$15,000,000
 - Up-Front New Equipment = \$7,500,000
 - Expected life of the new equipment is five years.
 - Depreciated at 1,500,000 p.a.
 - Housed in existing lab
 - Annual Overhead (SGaA Expenses) = \$2,800,000
 - Per Unit Cost = \$110so \$11,000,000 per year

Incremental Earnings Forecast

Revenue Estimates

- Sales = 100,000 units/year
- Per Unit Price = \$260.....so \$26,000,000 per year

Cost Estimates

- Up-Front R&D = \$15,000,000
- Up-Front New Equipment = \$7,500,000
- Per Unit Cost = \$110so \$11,000,000 per year
- Annual Overhead = \$2,800,000
- Expected life of the new equipment is five years.
- Depreciated at 1,500,000 p.a.
- Housed in existing lab

	Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)							
1 Sales		-	26,000	26,000	26,000	26,000	-
2 Cost of Goods Sold		-	(11,000)	(11,000)	(11,000)	(11,000)	-
3 Gross Profit		-	15,000	15,000	15,000	15,000	-
4 Selling, General, and Administrative		-	(2,800)	(2,800)	(2,800)	(2,800)	-
5 Research and Development		(15,000)	-	-	-	-	-
6 Depreciation		-	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7 EBIT		(15,000)	10,700	10,700	10,700	10,700	(1,500)
8 Income Tax at 20%		3,000	(2,140)	(2,140)	(2,140)	(2,140)	300
9 Unlevered Net Income		(12,000)	8,560	8,560	8,560	8,560	(1,200)

Tax credit

Capital Expenditures, Depreciation and Interest Expense

Depreciation

Interest Payments

- The \$7.5 million in new equipment is a cash expense, but it is not directly listed as an expense when calculating *earnings*. Instead, the firm deducts a fraction of the cost of these items each year as depreciation.

- Straight Line Depreciation

- The asset's cost is divided equally over its life. Annual Depreciation

$$= \frac{\$7.5 \text{ million}}{5 \text{ years}} = \$1.5 \frac{\text{million}}{\text{year}}$$

- Interest Payments

- In capital budgeting decisions, interest expense is typically not included.
 - The rationale is that the project should be judged on its own, not on how it will be financed.

Taxes

Definition:

Average tax rates are a measure of a business's tax burden;

Marginal rates measure the degree to which taxes affect economic incentives such as whether to invest or save more

Taxing Losses for Projects in Profitable Companies

Textbook Example 8.1

- Marginal Corporate Tax Rate
 - The tax rate on the marginal or **incremental** dollar of pre-tax income
 - Note: A negative tax is equal to a tax credit

$$\text{Income Tax} = \text{EBIT} \times \tau_c$$

- Unlevered Net Income Calculation

$$\begin{aligned}\text{Unlevered Net Income} &= \text{EBIT} \times (1 - \tau_c) \\ &= (\text{Revenues} - \text{Costs} - \text{Depreciation}) \times (1 - \tau_c)\end{aligned}$$

Example:

- Kellogg plans to launch a new line of high-fiber, gluten-free breakfast pastries. The heavy advertising expenses associated with the product launch will generate operating losses of \$20 million next year. Kellogg expects to earn pre-tax income of \$460 million from operations other than the new pastries next year. If Kellogg pays a 25% tax rate on its pre-tax income, **what will it owe in taxes next year without the new product? What will it owe with it?**



Textbook Example 8.1

Solution

Without the new pastries, Kellogg will owe
in corporate taxes next year.

With the new pastries, Kellogg's pre-tax income next year will be only

..... , and it will owe

..... in tax.

Thus, launching the new product reduces Kellogg's taxes next year by

..... , which equals

.....

Include forgone opportunities as implicit costs

Textbook Example 8.2

- Opportunity Cost
 - The value a resource could have provided in its best alternative use
 - In the HomeNet project example, space will be required for the investment. Even though the equipment will be **housed in an existing lab**, the opportunity cost of not using the space in an alternative way (e.g., renting it out) must be considered.

The Opportunity Cost of HomeNet's Lab Space

Problem

Suppose HomeNet's new lab will be housed in warehouse space that the company would have otherwise rented out for \$200,000 per year during years 1 – 4. How does this opportunity cost affect HomeNet's incremental earnings?

Solution

In this case, the opportunity cost of the warehouse space is the forgone rent. This cost would reduce HomeNet's incremental earnings during years 1-4 by

.....,

the after-tax benefit of renting out the warehouse space.

Account for how new projects affect existing business profits

Homenet Example

- Project Externalities
 - Indirect effects of the project that may affect the profits of other business activities of the firm. **Cannibalization** is when sales of a new product displaces sales of an existing product.
- Example:
 - For the cannibalization, suppose that the existing device wholesales for \$100 so the expected loss in sales is $25\% \times 100,000 \text{ units} \times \$100 \text{ unit} = \text{\$2.5 million}$ Compared to Table 8.1, the sales forecast falls from \$26 million to \$23.5 million. In addition, suppose the cost of the existing device is \$60 per unit.
 - Then, because Cisco will no longer need to produce as many of its existing product, the incremental cost of goods sold attributable to the HomeNet project is reduced by $25\% \times 100,000 \text{ units} \times (\$60 \text{ cost per unit}) = \text{\$1.5 million}$ from \$11 million to \$9.5 million. HomeNet's incremental gross profit therefore declines by $\text{\$2.5 million} - \text{\$1.5 million} = \text{\$1 million}$ once we account for this externality.
 - See Appendix for further details of this example
- Project Externalities
 - In the HomeNet project example, 25% of sales come from customers who would have purchased an existing Cisco wireless router if HomeNet were not available. Because this reduction in sales of the existing wireless router is a consequence of the decision to develop HomeNet, we must include it when calculating HomeNet's incremental earnings.

Ignore costs
already incurred—
they do not affect
future value

- **Sunk costs** are costs that have been or will be paid regardless of the decision whether or not the investment is undertaken
 - Sunk costs should not be included in the incremental earnings analysis
- Fixed Overhead Expenses
 - Typically overhead costs are fixed and not incremental to the project and should not be included in the calculation of incremental earnings.
- Past Research and Development Expenditures
 - Money that has already been spent on R&D is a **sunk cost** and therefore irrelevant.
 - The decision to continue or abandon a project should be based only on the incremental costs and benefits of the product going forward.
- Unavoidable Competitive Effects
 - When developing a new product, firms may be concerned about the cannibalization of existing products.
 - However, if sales are likely to decline in any case as a result of new products introduced by competitors, then these lost sales should be considered a sunk cost.

Textbook Example 8.3

Product Adoption and Price Changes

Problem

Suppose sales of HomeNet were expected to be 100,000 units in year 1, 125,000 units in years 2 and 3, and 50,000 units in year 4.

Suppose also that HomeNet's sale price and manufacturing cost are expected to decline by 10% per year, as with other networking products.

By contrast, selling, general, and administrative expenses are expected to rise with inflation by 4% per year.

Update the incremental earnings forecast in the spreadsheet in Table 8.2 to account for these effects.

Textbook Example 8.3

Solution

Home Net's incremental earnings with these new assumptions are shown in the spreadsheet below:

Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)						
1 Sales	—	23,500	26,438	23,794	8,566	—
2 Cost of Goods Sold	—	(9,500)	(10,688)	(9,619)	(3,463)	—
3 Gross Profit	—	14,000	15,750	14,175	5,103	—
4 Selling, General, and Administrative	—	(3,000)	(3,120)	(3,245)	(3,375)	—
5 Research and Development	(15,000)	—	—	—	—	—
6 Depreciation	—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7 EBIT	(15,000)	9,500	11,130	9,430	228	(1,500)
8 Income Tax at 20%	3,000	(1,900)	(2,226)	(1,886)	(46)	300
9 Unlevered Net Income	(12,000)	7,600	8,904	7,544	183	(1,200)

For example, sale prices in year 2 will be $\$260 \times 0.90 = \234 per unit for Home Net, and $\$100 \times 0.90 = \90 per unit for the cannibalized product.

Thus, incremental sales in year 2 are equal to $125,000 \text{ units} \times (\$234 \text{ per unit}) - 31,250 \text{ cannibalized units} \times (\$90 \text{ per unit}) = \$26.438 \text{ million}$.

*FCF = after-tax
operating profit +
depreciation –
capex – Δ NWC*

8.2 Calculating the Free Cash Flow from Earnings (1 of 3)

Definition:

The incremental effect of a project on a firm's available cash is its **free cash flow**.

Table 8.3

- Capital Expenditures and Depreciation
 - Capital Expenditures are the actual cash outflows when an asset is purchased
 - These cash outflows are included in calculating free cash flow
 - Depreciation is a non-cash expense
 - The free cash flow estimate is adjusted for this non-cash expense

	Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)							
1	Sales	—	23,500	23,500	23,500	23,500	—
2	Cost of Goods Sold	—	(9,500)	(9,500)	(9,500)	(9,500)	—
3	Gross Profit	—	14,000	14,000	14,000	14,000	—
4	Selling, General, and Administrative	—	(3,000)	(3,000)	(3,000)	(3,000)	—
5	Research and Development	(15,000)	—	—	—	—	—
6	Depreciation	—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7	EBIT	(15,000)	9,500	9,500	9,500	9,500	(1,500)
8	Income Tax at 20%	3,000	(1,900)	(1,900)	(1,900)	(1,900)	300
9	Unlevered Net Income	(12,000)	7,600	7,600	7,600	7,600	(1,200)
Free Cash Flow (\$000s)							
10	Plus: Depreciation	—	1,500	1,500	1,500	1,500	1,500
11	Less: Capital Expenditures	(7,500)	—	—	—	—	—
12	Less: Increases in NWC	—	(2,100)	—	—	—	2,100
13	Free Cash Flow	(19,500)	7,000	9,100	9,100	9,100	2,400

Calculating the Free Cash Flow from Earnings (2 of 3)

Table 8.4 Spreadsheet
HomeNet's Net Working Capital Requirements

- Net Working Capital (NWC)

$$\begin{aligned}\text{Net Working Capital} &= \text{Current Assets} - \text{Current Liabilities} \\ &= \text{Cash} + \text{Inventory} + \text{Receivables} - \text{Payables}\end{aligned}$$

– Most projects will require an investment in net working capital.

- **Trade credit** is the difference between receivables and payables.

– The increase in net working capital is defined as $\Delta NWC_t = NWC_t - NWC_{t-1}$

Year	0	1	2	3	4	5
Net Working Capital Forecast (\$000s)						
1 Cash Requirements	—	—	—	—	—	—
2 Inventory	—	—	—	—	—	—
3 Receivables (15% of Sales)	—	3,525	3,525	3,525	3,525	—
4 Payables (15% of COGS)	—	(1,425)	(1,425)	(1,425)	(1,425)	—
5 Net Working Capital	—	2,100	2,100	2,100	2,100	—

Textbook Example 8.4

Net Working Capital with Changing Sales

Problem

Forecast the required investment in net working capital for HomeNet under the scenario in Example 8.3. Assume no incremental inventory or cash requirements, but receivables (payables) 15% of sales (COGS)

Solution

Required investments in net working capital are shown below:

	Year	0	1	2	3	4	5
Net Working Capital Forecast (\$000s)							
1	Receivables (15% of Sales)	—	3,525	3,966	3,569	1,285	—
2	Payables (15% of COGS)	—	(1,425)	(1,603)	(1,443)	(519)	—
3	Net Working Capital	—	2,100	2,363	2,126	765	—
4	Increases in NWC	—	2,100	263	(237)	(1,361)	(765)

In this case, working capital changes each year. A large initial investment in working capital is required in year 1, followed by a small investment in year 2 as sales continue to grow. Working capital is recovered in years 3–5 as sales decline.

Calculating Free Cash Flow Directly

- Free Cash Flow

$$\text{Free Cash Flow} = \overbrace{(\text{Revenues} - \text{Costs} - \text{Depreciation}) \times (1 - \tau_c)}^{\text{Unlevered Net Income}} + \text{Depreciation} - \text{CapEx} - \Delta NWC \quad (\text{Eq. 8.5})$$

$$\text{Free Cash Flow} = (\text{Revenues} - \text{Costs}) \times (1 - \tau_c) - \text{CapEx} - \Delta NWC + \tau_c \times \text{Depreciation} \quad (\text{Eq. 8.6})$$

– The term $\tau_c \times \text{Depreciation}$ is called the depreciation tax shield

Calculating the NPV (1 of 2)

$$PV(FCF_t) = \frac{FCF_t}{(1+r)^t} = FCF_t \times \frac{1}{(1+r)^t}$$

t =year discount factor

- HomeNet NPV (WACC (weighted average cost of capital) = 12%)

$$NPV = -19,500 + \frac{7000}{1.12} + \frac{9100}{1.12^2} + \frac{9100}{1.12^3} + \frac{9100}{1.12^4} + \frac{2400}{1.12^5}$$

$$= \$7627$$

Table 8.5 Spreadsheet
Computing HomeNet's NP

Note: that the FCF line is based on Table 8.3

	Year	0	1	2	3	4	5
Net Present Value (\$000s)							
1 Free Cash Flow		(19,500)	7,000	9,100	9,100	9,100	2,400
2 Project Cost of Capital	12%						
3 Discount Factor		1.000	0.893	0.797	0.712	0.636	0.567
4 PV of Free Cash Flow		(19,500)	6,250	7,254	6,477	5,783	1,362
5 NPV		7,627					
6 IRR		27.9%					

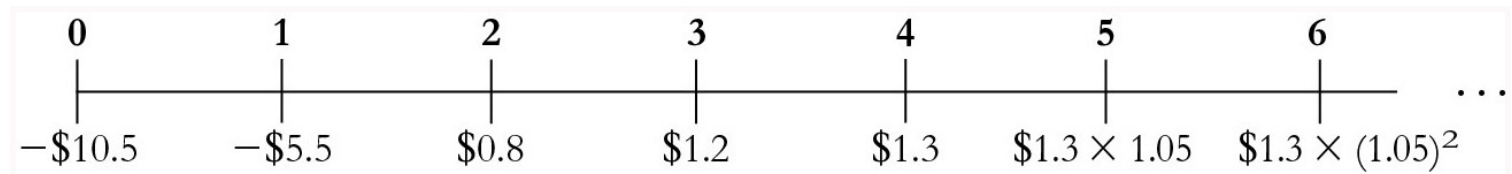
Terminal or Continuation Value

Textbook Example 8.7 (1 of 3)

- Terminal or Continuation Value
 - This amount represents the market value of the free cash flow from the project at all future dates.

Problem

Base Hardware is considering opening a set of new retail stores. The free cash flow projections for the new stores are shown below (in millions of dollars):



After year 4, Base Hardware expects free cash flow from the stores to increase at a rate of 5% per year. If the appropriate cost of capital for this investment is 10%, what continuation value in year 4 captures the value of future free cash flows in year 5 and beyond? **What is the NPV of the new stores?**

Textbook Example 8.7 (2 of 3)

Solution

Because the future free cash flow beyond year 4 is expected to grow at 5% per year, the continuation value in year 4 of the free cash flow in year 5 and beyond can be calculated as a constant growth perpetuity:

$$\begin{aligned}\text{Continuation Value in Year 4} &= PV(\text{FCF in Year 5 and Beyond}) \\ &= \\ &= \end{aligned}$$

Notice that under the assumption of constant growth, we can compute the continuation value as a multiple of the project's final free cash flow.

Textbook Example 8.7 (3 of 3)

We can restate the free cash flows of the investment as follows (in thousands of dollars):

Year	0	1	2	3	4
Free cash Flow (years 0–4)	(10,500)	(5,500)	800	1,200	1,300
Continuation value					?
Free cash flow	(10,500)	(5,500)	800	1,200	?

The NPV of the investment in the new stores is

$$NPV =$$

$$=$$

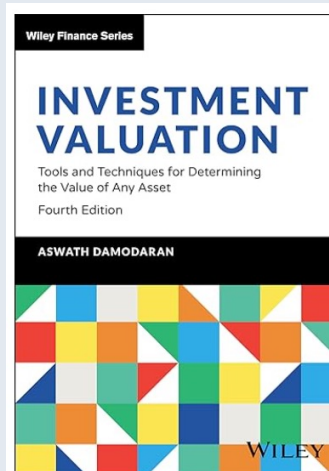
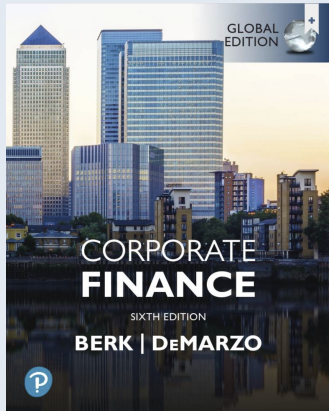
or million

Chapter Quiz

BD Chapter 8

1. Should you include sunk costs in the cash flow forecasts of a project? Why or why not?
2. Should you include opportunity costs in the cash flow forecasts of a project? Why or why not?
3. What adjustments must be made to a project's unlevered net income to determine its free cash flows?
4. How do you choose between mutually exclusive capital budgeting decisions?
5. What is the terminal value of a project?

Chapter 9 (BD)



Chapter Outline

- 9.1 The Dividend-Discount Model
- 9.2 Applying the Dividend-Discount Model
- 9.3 Total Payout and Free Cash Flow Valuation Models
- 9.4 Valuation Based on Comparable Firms

As **optional further reading** for those that want to learn more about advanced company valuation, I recommend

“Investment Valuation” (4th ed) by Aswath Damodaran

<https://pages.stern.nyu.edu/~adamodar/>

Chapter 9

Learning objectives

- Use the dividend-discount model to compute the value of a dividend-paying company's stock, whether the dividends grow at a constant rate starting now or at some time in the future.
- Discuss the determinants of future dividends and growth rate in dividends, and the sensitivity of the stock price to estimate those two factors.
- Given the retention rate and the return on new investment, calculate the growth rate in dividends, earnings, and share price.
- Assuming a firm has a long-term constant growth rate after time $N + 1$, use the constant growth model to calculate the terminal value of the stock at time N .
- Compute the stock value of a firm that pays dividends as well as repurchasing shares.
- Use the discounted free cash flow model to calculate the value of stock in a company with leverage.
- Use comparable firm multiples to estimate stock value.
- Explain why several valuation models are required to value a stock.

9.1 The Dividend-Discout Model

- A One-Year Investor
 - Potential Cash Flows
 - Dividend
 - Sale of Stock
 - Timeline for One-Year Investor



- Since the cash flows are risky, we must discount them at the **equity cost of capital**
- A One-Year Investor

$$P_0 = \left(\frac{Div_1 + P_1}{1 + r_E} \right)$$

Dividend Yields, Capital Gains, and Total Returns

$$r_E = \frac{Div_1 + P_1}{P_0} - 1 = \underbrace{\frac{Div_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

- Dividend Yield
- Capital Gain
 - Capital Gain Rate
- Total Return
 - Dividend Yield + Capital Gain Rate
 - The expected total return of the stock should equal the expected return of other investments available in the market with equivalent risk

Example: ETFs with high dividends

DIG DEEPER INTO DIVIDENDS

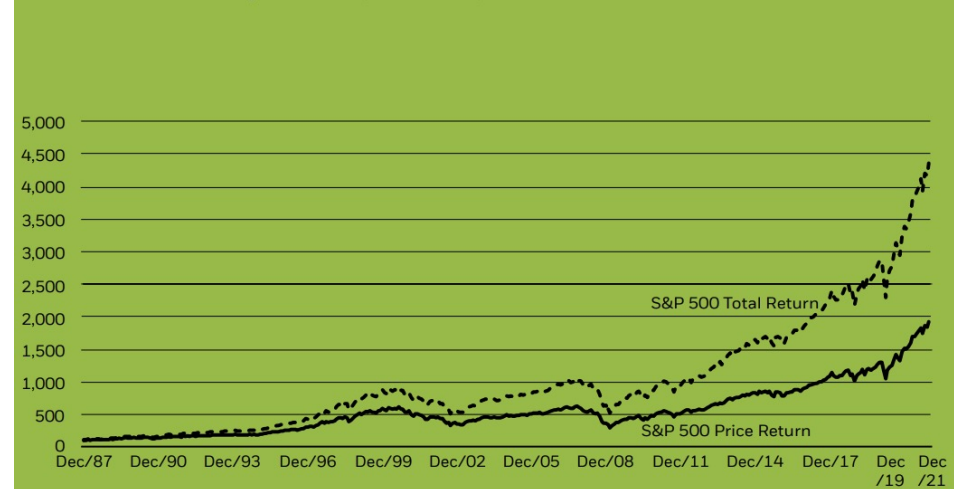
An iShares guide to dividend investing with ETFs



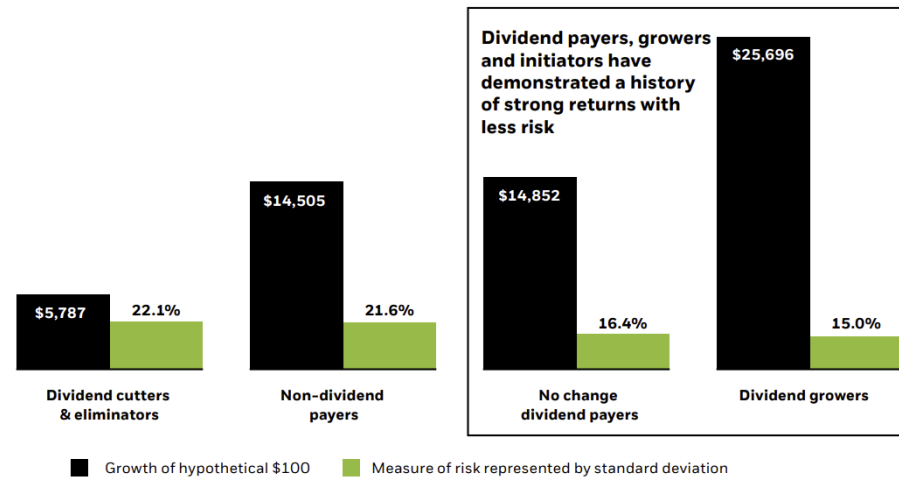
iShares
by BlackRock

Source: <https://www.ishares.com/us/literature/investor-guide/deep-dive-into-dividends-investing-guide-en.pdf>

S&P 500 Total return and price return (1987 - 2021)*



Risk and returns of the 500 largest U.S. stocks by dividend policy (1979 - 2021)*



Textbook Example 9.1

Stock Prices and Returns

Problem

Suppose you expect Walgreens Boots Alliance (a drugstore chain) to pay dividends of \$1.60 per share and trade for \$70 per share at the end of the year.

If investments with equivalent risk to Walgreen's stock have an expected return of 8.5%, what is the most you would pay today for Walgreen's stock?

What dividend yield and capital gain rate would you expect at this price?

Textbook Example 9.1 (2 of 2)

Solution

Using Eq. 9.1, we have

$$P_0 = \frac{Div_1 + p_1}{1 + r_E} = \text{_____} = \$ \dots \dots \dots$$

At this price, Walgreen's dividend yield is $\frac{Div_1}{P_0} = \text{_____} =$

The expected capital gain isper

share, for a capital gain rate of _____ =

Therefore, at this price, Walgreen's expected total return is ,
which is equal to its equity cost of capital.

The Dividend-Discount Model Equation

- What is the price if we plan on holding the stock for N years?

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \dots + \frac{Div_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$$

- This is known as the Dividend-Discount Model

- Note that the above equation (9.4) holds for any horizon N

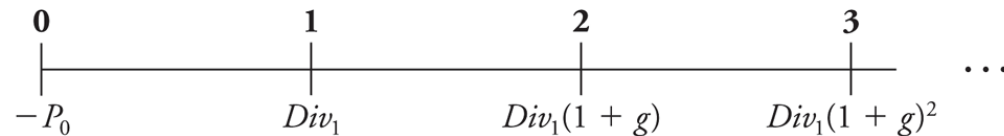
- Thus all investors (with the same beliefs) will attach the same value to the stock, independent of their investment horizons

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \frac{Div_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{Div_n}{(1 + r_E)^n}$$

- The price of any stock is equal to the present value of the expected future dividends it will pay

9.2 Applying the Discount-Dividend Model (1 of 2)

- Constant Dividend Growth
 - The simplest forecast for the firm's future dividends states that they will grow at a constant rate, g , forever



- Constant Dividend Growth Model

$$P_0 = \frac{Div_1}{r_E - g}$$

$$r_E = \frac{Div_1}{P_0} + g$$

- The value of the firm depends on the current dividend level, the cost of equity, and the growth rate

Textbook Example

9.2 (1 of 2)

Valuing a Firm with Constant Dividend Growth

Problem

Consolidated Edison, Inc. (Con Edison), is a regulated utility company that services the New York City area.

Suppose Con Edison plans to pay \$3.40 per share in dividends in the coming year.

If its equity cost of capital is 6% and dividends are expected to grow by 2% per year in the future, estimate the value of Con Edison's stock.

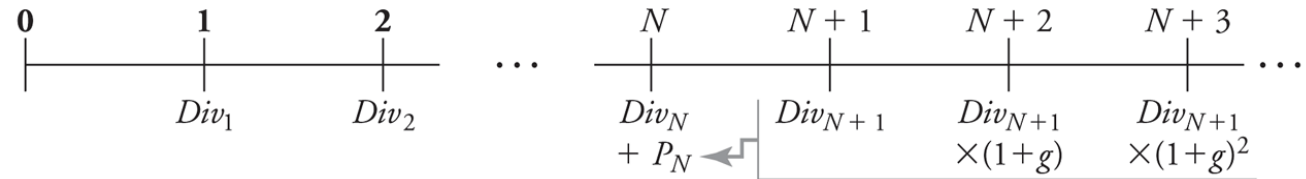
Textbook Example 9.2 (2 of 2)

Solution

If dividends are expected to grow perpetually at a rate of 2% per year, we can use Eq. 9.6 to calculate the price of a share of Con Edison stock:

$$P_0 = \frac{\text{Div}_1}{r_E - g} = \frac{\text{Div}_1}{0.10 - 0.02} = \$$$

Changing Growth Rates (3 of 3)



$$P_N = \frac{Div_{N+1}}{r_E - g}$$

- Dividend-Discount Model with Constant Long-Term Growth

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \dots + \frac{Div_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{Div_{N+1}}{r_E - g} \right)$$

Textbook Example 9.5 (1 of 3)

Valuing a Firm with Two Different Growth Rates

- **Problem**

- Small Fry, Inc., has just invented a potato chip that looks and tastes like a French fry.
- Given the phenomenal market response to this product, Small Fry is reinvesting all of its earnings to expand its operations.
- Earnings were \$2 per share this past year and are expected to grow at a rate of 20% per year until the end of year 4.
- At that point, other companies are likely to bring out competing products.
- Analysts project that at the end of year 4, Small Fry will cut investment and begin paying 60% of its earnings as dividends and its growth will slow to a long-run rate of 4%.
- If Small Fry's equity cost of capital is 8%, what is the value of a share today?

Textbook Example 9.5

Solution

We can use Small Fry's projected earnings growth rate and payout rate to forecast its future earnings and dividends as shown in the following spreadsheet:

	Year	0	1	2	3	4	5	6
Earnings								
1	EPS Growth Rate (versus prior year)		20%	20%	20%	20%	4%	4%
2	EPS	\$2.00	\$2.40	\$2.88	\$3.46	\$4.15	\$4.31	\$4.49
Dividends								
3	Dividend Payout Rate		0%	0%	0%	60%	60%	60%
4	Dividend		\$ —	\$ —	\$ —	\$2.49	\$2.59	\$2.69

Starting from \$2.00 in year 0, EPS grows by 20% per year until year 4, after which growth slows to 4%. Small Fry's dividend payout rate is zero until year 4, when competition reduces its investment opportunities and its payout rate rises to 60%. Multiplying EPS by the dividend payout ratio, we project Small Fry's future dividends in line 4.

From year 4 onward, Small Fry's dividends will grow at the expected long-run rate of 4% per year. Thus, we can use the constant dividend growth model to project Small Fry's share price at the end of year 3. Given its equity cost of capital of 8%,

$$P_3 = \frac{\text{Div}_4}{r_E - g} = \frac{\$2.49}{0.08 - 0.04} = \$62.25$$

We then apply the dividend-discount model (Eq. 9.4) with this terminal value:

$$P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \frac{\text{Div}_3}{(1 + r_E)^3} + \frac{P_3}{(1 + r_E)^3} = \frac{\$62.25}{(1.08)^3} = \$49.42$$

As this example illustrates, the dividend-discount model is flexible enough to handle any forecasted pattern of dividends.

The Discounted Free Cash Flow Model

- Discounted Free Cash Flow Model
 - Determines the value of the firm to all investors, including both equity and debt holders

$$\text{Enterprise Value} = \text{Market Value of Equity} + \text{Debt} - \text{Cash}$$

- The enterprise value can be interpreted as the net cost of acquiring the firm's equity, taking its cash, paying off all debt, and owning the unlevered business

- Valuing the Enterprise

Free Cash Flow

$$= \overbrace{\text{EBIT} \times (1 - \tau_c)}^{\text{Unlevered Net Income}} + \text{Depreciation} - \text{Capital Expenditures} - \text{Increases in Net Working Capital}$$

- Free Cash Flow

- Cash flow available to pay both debt holders and equity holders

- Discounted Free Cash Flow Model

$$V_0 = \text{PV}(\text{Future Free Cash Flow of Firm})$$

$$P_0 = \frac{V_0 + \text{Cash}_0 - \text{Debt}_0}{\text{Shares Outstanding}_0}$$

Eq (9.22)

The Discounted Free Cash Flow Model

- Implementing the Model
 - Since we are discounting cash flows to both equity holders and debt holders, the free cash flows should be discounted at the firm's weighted average cost of capital, r_{wacc} . If the firm has no debt, $r_{wacc} = r_E$

$$V_0 = \frac{FCF_1}{1 + r_{wacc}} + \frac{FCF_2}{(1 + r_{wacc})^2} + \dots + \frac{FCF_N}{(1 + r_{wacc})^N} + \frac{V_N}{(1 + r_{wacc})^N}$$

- Often, the terminal value is estimated by assuming a constant long-run growth rate g_{FCF} for free cash flows beyond year N, so that

$$V_N = \frac{FCF_{N+1}}{r_{wacc} - g_{FCF}} = \left(\frac{1 + g_{FCF}}{(r_{wacc} - g_{FCF})} \right) \times FCF_N \quad \text{Eq (9.24)}$$

Textbook Example 9.7 (1 of 3)

2005 Kenneth Cole ad



Valuing Kenneth Cole Using Free Cash Flow

Problem

- Kenneth Cole (KCP) had sales of \$518 million in 2005. Suppose you expect its sales to grow at a 9% rate in 2006, but that this growth rate will slow by 1% per year to a long –run growth rate for the apparel industry of 4% by 2011.
- Based on KCP's past profitability and investment needs, you expect EBIT to be 9% of sales, increases in net working capital requirements to be 10% of any increase in sales, and net investment (capital expenditures in excess of depreciation) to be 8% of any increase in sales.
- If KCP has \$100 million in cash, \$3 million in debt, 21 million shares outstanding, a tax rate of 37%, and a weighted average cost of capital of 11%, **what is your estimate of the value of KCP's stock in early 2006?**

Textbook Example 9.7 (2 of 3)

Solution

Using Eq. 9.20, we can estimate KCP's future free cash flow based on the estimates above as follows:

	Year	2005	2006	2007	2008	2009	2010	2011
FCF Forecast (\$ millions)								
1	Sales	518.0	564.6	609.8	652.5	691.6	726.2	755.3
2	Growth versus Prior Year		9.0%	8.0%	7.0%	6.0%	5.0%	4.0%
3	EBIT (9% of sales)		50.8	54.9	58.7	62.2	65.4	68.0
4	Less: Income Tax (37% EBIT)		(18.8)	(20.3)	(21.7)	(23.0)	(24.2)	(25.1)
5	Less: Net Investment (8% ΔSales)		(3.7)	(3.6)	(3.4)	(3.1)	(2.8)	(2.3)
6	Less: Inc. in NWC (10% ΔSales)		(4.7)	(4.5)	(4.3)	(3.9)	(3.5)	(2.9)
7	Free Cash Flow		23.6	26.4	29.3	32.2	35.0	37.6

FCF₂₀₁₁

Because we expect KCP's free cash flow to grow at a constant rate after 2011, we can use Eq. 9.24 to compute a terminal enterprise value:

Textbook Example 9.7 (3 of 3)

$$V_{2011} = \left(\frac{1 + g_{\text{FCF}}}{r_{\text{wacc}} - g_{\text{FCF}}} \right) \times \text{FCF}_{2011} = \underline{\hspace{2cm}}$$

From Eq. 9.23, KCP's current enterprise value is the present value of its free cash flows plus the terminal enterprise value:

$$\begin{aligned} V_0 &= \frac{23.6}{1.11} + \frac{26.4}{1.11^2} + \frac{29.3}{1.11^3} + \frac{32.2}{1.11^4} + \frac{35.0}{1.11^5} + \frac{37.6}{1.11^6} + \frac{?}{1.11^6} \\ &= 126.1964 + \frac{?}{1.11^6} \end{aligned}$$

$$= \$ \underline{\hspace{2cm}} \text{ million}$$

We can now estimate the value of a share of KCP's stock using Eq. 9.22:

$$P_0 = \underline{\hspace{2cm}}$$

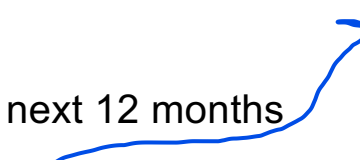
The Discounted Free Cash Flow Model (5 of 5)

- Connection to Capital Budgeting
 - The firm's free cash flow is equal to the sum of the free cash flows from the firm's current and future investments, so we can interpret the firm's enterprise value as the total NPV that the firm will earn from continuing its existing projects and initiating new ones.
 - The NPV of any individual project represents its contribution to the firm's enterprise value. To maximize the firm's share price, we should accept projects that have a positive NPV.

9.4 Valuation Based on Comparable Firms

...

- Method of Comparables (Comps)
 - Estimate the value of the firm based on the value of other, comparable firms or investments that we expect will generate very similar cash flows in the future. **Yahoo.finance Example**
- Valuation Multiple
 - A ratio of firm's value to some measure of the firm's scale or cash flow
- The Price-Earnings Ratio
 - *P/E* Ratio
 - Share price divided by earnings per share
- Trailing Earnings
 - Earnings over the last 12 months
- Trailing *P/E*
- Forward Earnings
 - Expected earnings over the next 12 months
- Forward *P/E*

$$\text{Forward P/E} = \frac{P_0}{\text{EPS}_1} = \frac{\text{Div}_1 / \text{EPS}_1}{r_E - g} = \frac{\text{Dividend Payout Rate}}{r_E - g}$$


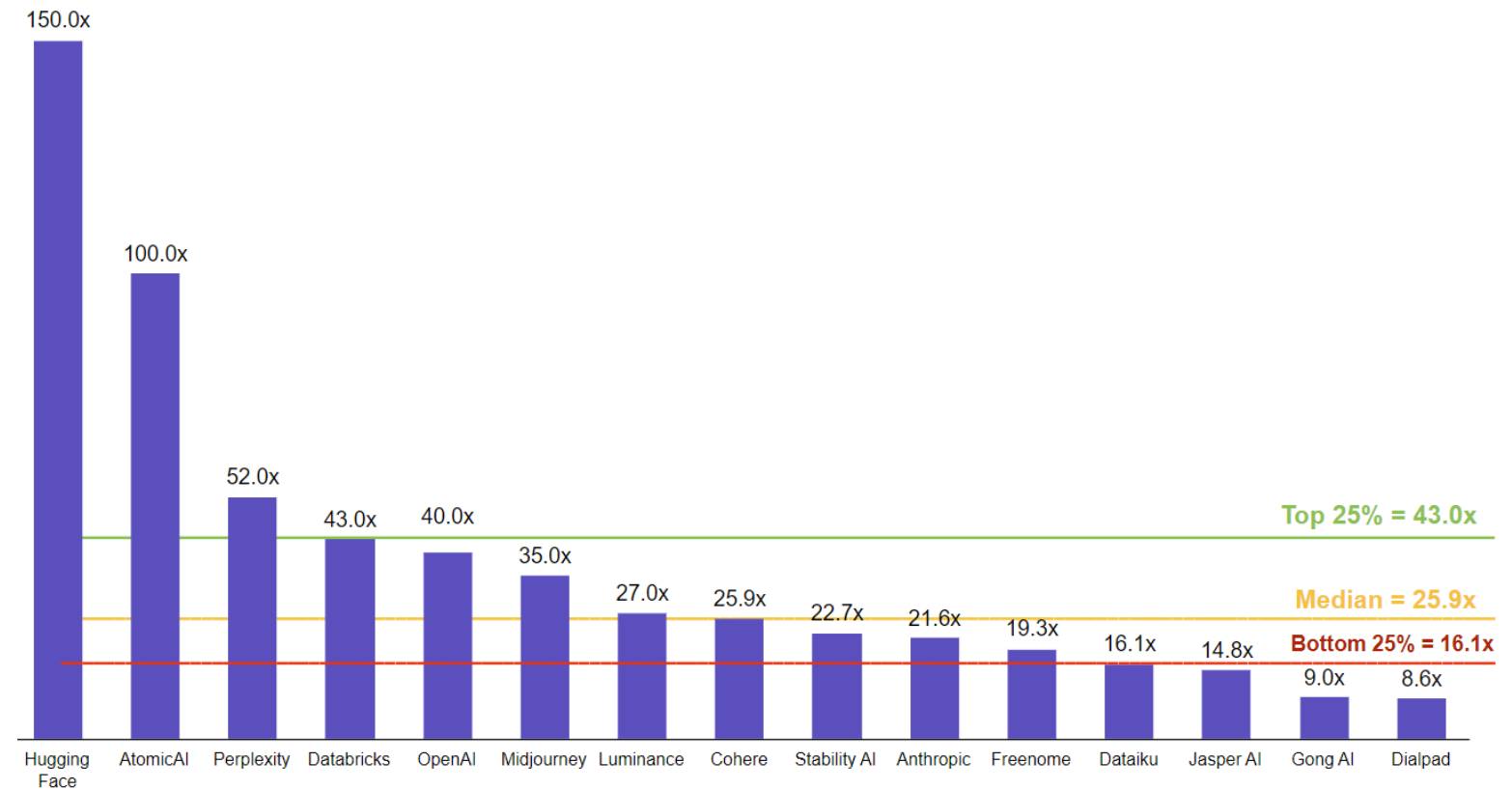
$$\frac{\text{Div}_1}{r - g}$$

Recent example of a valuation multiple i...

To the right, are a set of EV/revenue multiples from large capital raising transactions and M&A deals.

The median revenue multiple for AI companies stood at 25.8x

As AI is a relatively new, as well as capital-intensive segment of technology, there majority of the deals are concentrated in capital raisings rather than traditional exits.



Example

Valuation Using the Price-Earnings Ratio

Problem

Suppose furniture manufacturer Herman Miller, Inc., has earnings per share of \$1.99.

If the average P/E of comparable furniture stocks is 24.6, estimate a value for Herman Miller using the P/E as a valuation multiple.

What are the assumptions underlying this estimate?

Example

Solution

Solution

We estimate a share price for Herman Miller by multiplying its EPS by the *P/E* of comparable firms.

Thus, $P_0 = \$1.99 \times 24.6 = \48.95 .

This estimate assumes that Herman Miller will have similar future risk, payout rates, and growth rates to comparable firms in the industry.

Valuation Multiples (4 of 5)

Textbook Example 9.10

- Enterprise Value Multiples

$$\frac{V_0}{\text{EBITDA}_1} = \frac{\text{FCF}_1 / \text{EBITDA}_1}{r_{\text{wacc}} - g_{\text{FCF}}}$$

- This valuation multiple is higher for firms with high growth rates and low capital requirements (so that free cash flow is high in proportion to EBITDA)

Valuation Using an Enterprise Value Multiple

Problem

Suppose Rocky Shoes and Boots (RCKY) has earnings per share of \$2.30 and EBITDA of \$30.7 million. RCKY also has 5.4 million shares outstanding and debt of \$125 million (net of cash).

You believe Deckers Outdoor Corporation is comparable to RCKY in terms of its underlying business, but Deckers has little debt.

If Deckers has a P/E of 13.3 and an enterprise value to EBITDA multiple of 7.4, estimate the value of RCKY's shares using both multiples. Which estimate is likely to be more accurate?

Textbook Example 9.10 (2 of 2)

Solution

Using Decker's P/E, we would estimate a share price for RCKY of

$$P_0 = 2.3 \times 13.3$$

Using the enterprise value to EBITDA multiple, we would estimate RCKY's enterprise value to be

$$V_0 = 30.7 \times 7.4$$

We then subtract debt and divide by the number of shares to estimate RCKY's share price:

$$P_0 = \frac{V_0 + \text{Cash} - \text{Debt}}{\# \text{ (number)}} = 18.93$$

Because of the large difference in leverage between the firms, we would expect the second estimate, which is based on enterprise value, to be more reliable.

Valuation Multiples

(5 of 5)

- Other Multiples
 - Multiple of sales
 - Price to book value of equity per share
 - Enterprise value per subscriber
 - Used in cable TV industry
- Limitations of Multiples
 - When valuing a firm using multiples, there is no clear guidance about how to adjust for differences in expected future growth rates, risk, or differences in accounting policies
 - Comparables only provide information regarding the value of a firm relative to other firms in the comparison set
 - Using multiples will not help us determine if an entire industry is overvalued
- Comparison with Discounted Cash Flow Methods
- Discounted cash flows methods have the advantage that they can incorporate specific information about the firm's cost of capital or future growth
 - The discounted cash flow methods have the potential to be more accurate than the use of a valuation multiple

Table 9.1 Stock Prices and Multiples for the Footwear Industry, January 2006

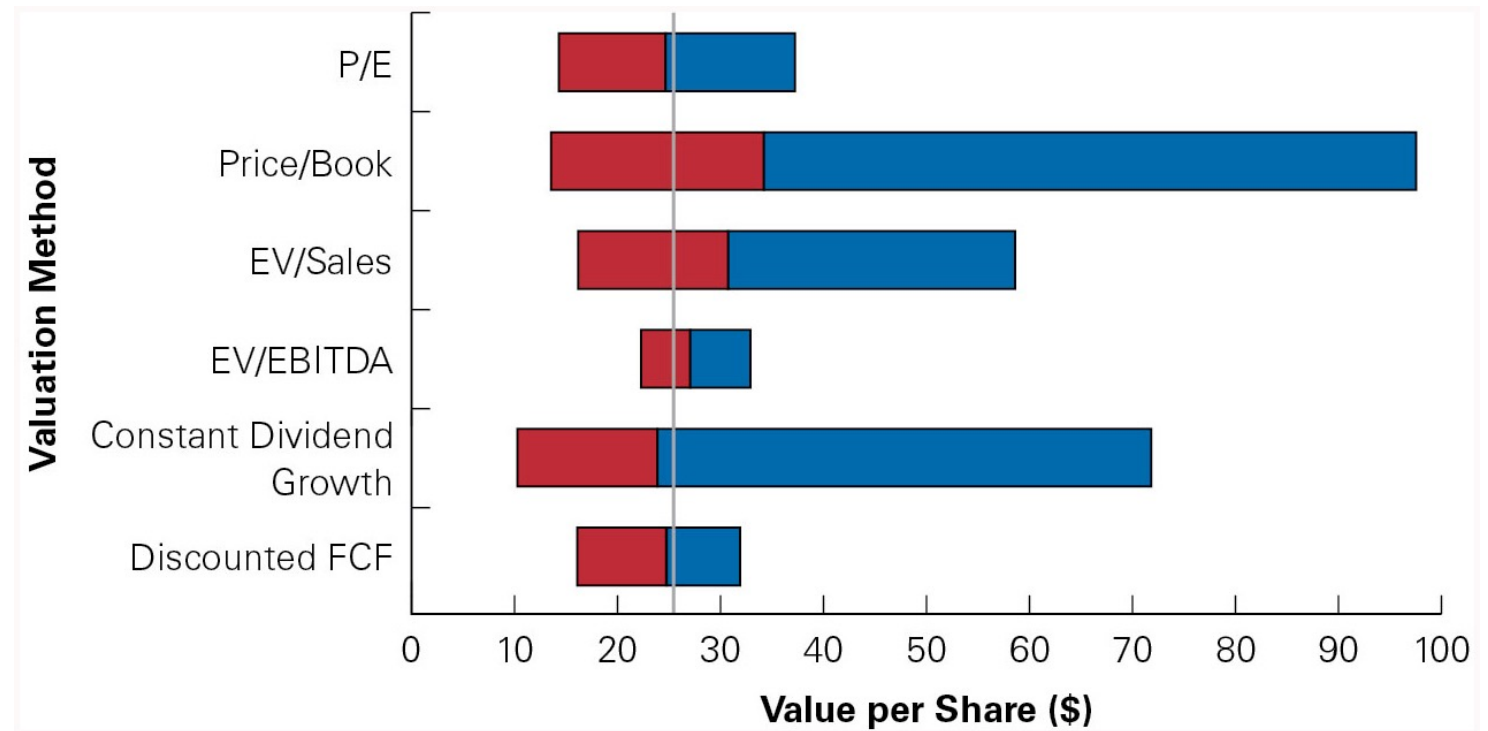
.... Not a boring industry in 2023 as Adidas' fate shows....

Stock Valuation Techniques:
The Final Word

Ticker	Name	Stock Price (\$)	Market Capitalization (\$ millions)	Enterprise Value (\$ millions)	P/E	Price/Book	Enterprise Value/Sales	Enterprise Value/EBITDA
KCP	Kenneth Cole Productions	26.75	562	465	16.21	2.22	0.90	8.36
NKE	NIKE, Inc.	84.20	21,830	20,518	16.64	3.59	1.43	8.75
PMMAY	Puma AG	312.05	5,088	4,593	14.99	5.02	2.19	9.02
RBK	Reebok International	58.72	3,514	3,451	14.91	2.41	0.90	8.58
WWW	Wolverine World Wide	22.10	1,257	1,253	17.42	2.71	1.20	9.53
BWS	Brown Shoe Company	43.36	800	1,019	22.62	1.91	0.47	9.09
SKX	Skechers U.S.A.	17.09	683	614	17.63	2.02	0.62	6.88
SRR	Stride Rite Corp.	13.70	497	524	20.72	1.87	0.89	9.28
DECK	Deckers Outdoor Corp.	30.05	373	367	13.32	2.29	1.48	7.44
WEYS	Weyco Group	19.90	230	226	11.97	1.75	1.06	6.66
RCKY	Rocky Shoes & Boots	19.96	106	232	8.66	1.12	0.92	7.55
DFZ	R.G. Barry Corp.	6.83	68	92	9.20	8.11	0.87	10.75
BOOT	LaCrosse Footwear	10.40	62	75	12.09	1.28	0.76	8.30
Average (excl. KCP)					15.01	2.84	1.06	8.49
Max (relative to Avg.)					+51%	+186%	+106%	+27%
Min (relative to Avg.)					-42%	-61%	-56%	-22%

- No single technique provides a final answer regarding a stock's true value
- All approaches require assumptions or forecasts that are too uncertain to provide a definitive assessment of the firm's value
 - Most real-world practitioners use a combination of these approaches and gain confidence if the results are consistent across a variety of methods

Figure 9.2 Range of Valuations for KCP Stock Using Alternative Valuation Methods



Chapter 9 Quiz

1. What discount rate do you use to discount the future cash flows of a stock?
2. Does an investor's expected holding period affect the amount they would be willing to pay for a stock?
3. How can a firm increase its future dividend per share?
4. What is the enterprise value of a firm?
5. What are the implicit assumptions made when valuing a firm using multiples?

Main Takeaways (1/2)

Week 3

1. Because dividend cash flows are risky, dividends must be discounted at the equity cost of capital, which is the expected return of other securities available in the market with equivalent risk to the firm's equity.
2. If the firm undertakes share repurchases, it is more reliable to use the total payout model to value the firm. In this model, the value of equity equals the present value of future total dividends and repurchases
3. The firm's enterprise value (the market value of equity plus debt, less excess cash) equals the present value of the firm's future free cash flow.
4. We discount cash flows using the weighted average cost of capital, which is the expected return the firm must pay to investors to compensate them for the risk of holding the firm's debt and equity together.
5. We can estimate a terminal enterprise value by assuming free cash flow grows at a constant rate (typically equal to the rate of long-run revenue growth).
6. We can also value stocks by using valuation multiples based on comparable firms. Multiples commonly used for this purpose include the P/E ratio and the ratio of enterprise value to EBITDA. Using multiples assumes that comparable firms have the same risk and future growth as the firm being valued. No valuation model provides a definitive value for the stock. It is best to use several methods to identify a reasonable range for the value.

Main Takeaways (2/2)

Week 3

1. We use the NPV rule to evaluate capital budgeting decisions, making decisions that maximize NPV. When deciding to accept or reject a project, we accept projects with a positive NPV.
2. The incremental earnings of a project comprise the amount by which the project is expected to change the firm's earnings.
3. We compute free cash flow from incremental earnings by eliminating all non-cash expenses and including all capital investment.
 1. Depreciation is not a cash expense, so it is added back.
 2. Actual capital expenditures are deducted.
 3. Increases in net working capital are deducted.
4. The discount rate for a project is its cost of capital: The expected return of securities with comparable risk and horizon.

Appendix Indirect Effects on Incremental Earnings (2 of 4)

Homenet Example

Table 8.1

	Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)							
1 Sales		-	26,000	26,000	26,000	26,000	-
2 Cost of Goods Sold		-	(11,000)	(11,000)	(11,000)	(11,000)	-
3 Gross Profit		-	15,000	15,000	15,000	15,000	-
4 Selling, General, and Administrative		-	(2,800)	(2,800)	(2,800)	(2,800)	-
5 Research and Development		(15,000)	-	-	-	-	-
6 Depreciation		-	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7 EBIT		(15,000)	10,700	10,700	10,700	10,700	(1,500)
8 Income Tax at 20%		3,000	(2,140)	(2,140)	(2,140)	(2,140)	300
9 Unlevered Net Income		(12,000)	8,560	8,560	8,560	8,560	(1,200)

Table 8.2

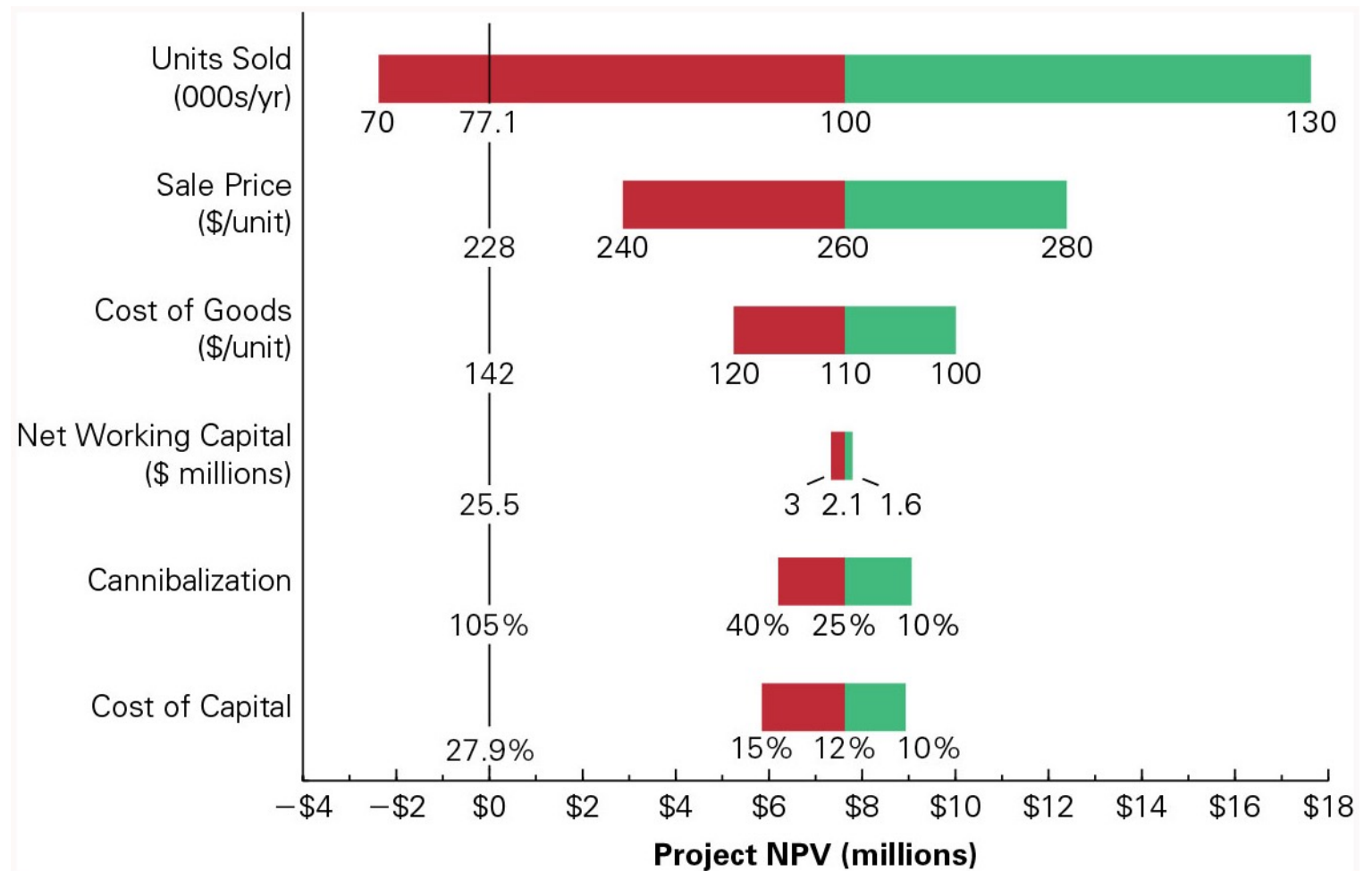
	Year	0	1	2	3	4	5
Incremental Earnings Forecast (\$000s)							
1 Sales		—	23,500	23,500	23,500	23,500	—
2 Cost of Goods Sold		—	(9,500)	(9,500)	(9,500)	(9,500)	—
3 Gross Profit		—	14,000	14,000	14,000	14,000	—
4 Selling, General, and Administrative		—	(3,000)	(3,000)	(3,000)	(3,000)	—
5 Research and Development		(15,000)	—	—	—	—	—
6 Depreciation		—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7 EBIT		(15,000)	9,500	9,500	9,500	9,500	9,500
8 Income Tax at 20%		3,000	(1,900)	(1,900)	(1,900)	(1,900)	300
9 Unlevered Net Income		(12,000)	7,600	7,600	7,600	7,600	(1,200)

Appendix: Sensitivity Analysis

Definition:

Sensitivity analysis breaks the NPV calculation into its component assumptions and shows how the NPV varies as the underlying assumptions change.

Figure 8.1 HomeNet's NPV Under Best-and Worst-Case Parameter Assumptions



Any questions?

Appendix

Internal Rate of return

Textbook Example 4.16 (1 of 2)

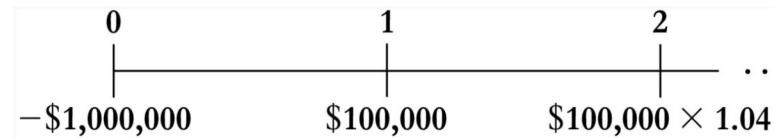
- In some situations, you know the present value and cash flows of an investment opportunity, but you do not know the **internal rate of return (IRR)**, the interest rate that sets the net present value of the cash flows equal to zero.
- **Problem: Computing the IRR for a Perpetuity**
 - Jessica has just graduated with her MBA. Rather than take the job she was offered at a prestigious investment bank—Baker, Bellingham, and Botts—she has decided to go into business for herself.
 - She believes that her business will require an initial investment of \$1 million. after that, it will generate a cash flow of \$100,000 at the end of one year, and this amount will grow by 4% per year thereafter.
 - What is the IRR of this investment opportunity?

Internal Rate of return

Textbook Example 4.16 (2 of 2)

Solution

The timeline is



The timeline shows that the future cash flows are a growing perpetuity with a growth rate of 4%.

Recall from Eq. 4.11 that the PV of a growing perpetuity is $\frac{C}{(r - g)}$.

Thus, the NPV of this investment would equal zero if $1,000,000 = \frac{100,000}{r - 0.04}$

We can solve this equation for r

$$r = \frac{100,000}{1,000,000} + 0.04 = 0.14$$

So, the IRR on this investment is 14%.